Sarah Thomas

+91-9993137915 | sarahtthomas29@gmail.com | linkedin.com/in/sarah-thomas-40a301289 | github.com/THOMAS-SARAH

Education

Vellore Institute of Technology September 2023 – May 2027
B.Tech in Computer Science – Specialization in Cloud Computing and Automation *CGPA: 8.81/10*

St. Joseph's Convent Sr. Sec. School, Sagar, Madhya Pradesh March 2022 – May 2023
All India Senior School Certificate Examination *CGPA: 7.88/10*

St. Joseph's Convent Sr. Sec. School, Sagar, Madhya Pradesh March 2020 – June 2021
All India Secondary School Examination *CGPA: 9.2/10*

Experience

MPOnline Limited (Govt. of Madhya Pradesh & TCS Joint Venture) 20 May 2026 – 31 Aug 2026
Software Engineering & Operations Intern *Remote*

- Owned end-to-end execution of database architectures and backend modules, driving features from initial user problem identification straight to production under Agile workflows.
- Collaborated in a 7-member team to map logistics workflows and build an AI-Powered Smart Logistics System; translated field operational requirements into real-time tracking features using ASP.NET Core and SignalR.
- Automated fragmented operational workflows by integrating generative AI and n8n automation pipelines, eliminating manual bottlenecks for internal team tasks.

Technical Skills

Cloud & DevOps: AWS (Lambda, S3, API Gateway, EC2, IAM, CloudWatch), Docker, Docker Compose, Prometheus, Terraform, GitHub Actions, Git, GitHub, Boto3

Languages & Frameworks: Java, Python, Linux/Bash, FastAPI, REST APIs, AIOps, Scikit-Learn, OpenCV, ASP.NET Core, C#

Databases & Tools: MongoDB, Microsoft SQL Server, OpenAI API, Pandas, Matplotlib, EF Core, SignalR, n8n

Projects

AI Incident & Log Analysis Assistant

Python, FastAPI, OpenAI API *github.com/THOMAS-SARAH/exception_analyzer*

- Shipped a live operational microservice that processes messy system logs to automatically generate structured, actionable Root-Cause Analysis (RCA) reports.
- Designed the tool to act as a force multiplier for technical operations teams, turning confusing system telemetry errors into clear, step-by-step remediation tasks.

AI-Driven Cloud Resource & Cost Optimizer

Prometheus, Terraform, AWS, FastAPI *github.com/THOMAS-SARAH/multi-region-aws-telemetry-pipeline*

- Built and deployed an active infrastructure pipeline that continuously monitors real-time CPU/memory utilization and flags complex system anomalies.
- Solved real-world cost inefficiencies by provisioning multi-region infrastructure via modular Terraform templates and building automated backend endpoints to self-optimize resource allocation.

Hybrid Ensemble Model for Enhanced Cancer Cell Classification

Python, Scikit-Learn, OpenCV, Pandas, Matplotlib *github.com/THOMAS-SARAH/CANCER_CELL_DETECTION*

- Developed a hybrid ensemble learning model combining multiple classifiers to distinguish cancerous cells, optimizing data preprocessing and extraction pipelines via OpenCV.
- Evaluated complex model performance using accuracy and AUC metrics, visualizing comparative validation results with Pandas and Matplotlib.

Certifications and Extra-Curricular

Certifications: ServiceNow University Virtual Internship Program; NPTEL – Internet Of Things, Cloud Computing and Distributed Systems; AWS Educate – Introduction to Generative AI.

Extra-Curricular & Leadership:

- **Education Minister, School Council:** Managed cross-functional coordination and gathered requirements directly from students and faculty to pitch educational initiatives to school leadership.
- **Club Coordinator, E-Green Club:** Led regular alignment reviews, communicated project progress and blockers, and integrated feedback from members to execute community initiatives.